

Experiment No. 1

Topic: Agile Requirements Gathering with User Stories (For Functional Requirement)

Aim: To identify functional requirement for any software development project.

Learning Outcome: *Study consists of learning the following concepts*

- Write effective User stories for the functional requirements of selected projects.
- Acceptance criteria and Definition of done.
- Maintain different versions of requirement documents.

Part A:

Theory :

Agile Requirements Gathering with User Stories

Just as in traditional projects, Agile projects start with basic requirements gathering. Knowing the goals of your end users and project stakeholders is the necessary first step to any successful project.

However, instead of trying to nail down all the details up front, your goal in an Agile project is to capture just enough information to have a conversation with the customer at a later date. You capture this information as "user stories" -- short descriptions of the functionality in customer terms.

A user story is a placeholder and a promise to have a future conversation about the needed functionality. By delaying the detailed conversation until shortly before development, you can avoid some of the waste that is typical of projects where detailed requirements are gathered early but become invalid before the work begins.

In addition, you may discover that certain requested features are no longer needed by the customer after a few months, and you'll have saved the cost and effort of a detailed analysis.

A useful structure for brainstorming user stories (although certainly not the only good one) is as follows:

As a [Type of User], [Function to Perform] so that [Business Value]

In practice, you might not actually write your stories exactly this way, but keeping this format in mind helps focus the story descriptions so that they stay customer and business value oriented instead of technical in nature.

For example, in a sales management system, some typical user stories might be:

- As a salesperson, I want to add a new contact so that I can follow up later with prospects.
- As a sales manager, view new contacts added by salesperson, so I can track leads

- As a system administrator, add a new salesperson so they can access the system

At the start of a project, you'll want to gather the key project stakeholders to brainstorm possible user stories at this level of detail. By elaborating all of the possible users of the system first, you can then determine what functionality each type of user might need.

It's perfectly fine if some of the user stories aren't uncovered in this first planning session. The key strength of Agile development approaches is the ability to incorporate new requests for functionality as they are discovered. This takes a great deal of pressure off of your customers, since they know they can change their minds if the business climate changes or if they forget something important.

Remember that the point is not to decide about how these stories can be implemented at this stage, it's just a first cut at understanding the scope of the system. Later, you'll prioritize these stories with your customers, and select a subset of these for your first release. You'll then choose an even smaller subset for your first iteration (a fixed length time period during which development will take place, usually 1 to 4 weeks).

It's during the iteration planning process that you'll have the more detailed conversations with your customers that clarify the issues. And even during development, it's critical to stay in close contact with the customers to be able to ask questions.

Format for Writing User Stories

A user story typically follows a simple template:

As a <type of user>, I want <to perform some task> so that I can <achieve some goal/benefit/value>

Few Examples

- As a customer, I want to withdraw cash from an ATM So that I don't have to wait in line at the bank.
- Acceptance Criterion 1:
 - Given that the account is credit worthy ○ the card is valid
 - The dispenser contains cash When the customer requests the cash ○ Then ensure the account is debited ○ ensure cash dispensed
 - ✦ Ensure the card is returned.
- Acceptance Criterion 2:
 - Given that the account is overdrawn
 - The card is valid, When the customer requests the cash Then ensure the rejection message is displayed
 - And ensure cash is not dispensed.

Another sample user story

Story name

Value statement

Attach details and documents when necessary

What is required for the business and product owner to accept the story

What is required by the team (quality/standards) before sending out for review. Does not change from one story to another. Mature teams may post this on the wall of the team working area instead of within each story.

Size (effort) estimate, in relative points

The screenshot shows a web application for editing a user story. The form is titled 'Edit User Story » US9: Credit card payments'. It has a 'General' tab selected. The form fields are as follows:

- ID:** US9
- Name:** Credit card payments
- Tags:** Choose Tags
- Description:** As a purchaser on the website, I want the ability to pay with a credit card, So that I may immediately confirm my purchase.
- Acceptance Criteria:**
 - Accept Discover, Visa, MC
 - Validate CC# when entered
 - Validate expiration date and CVV
 - Validate billing address
 - Generate success and failure messages after processing
- Definition of Done:**
 - Passes all regression tests
 - Passes testing per acceptance criteria items
 - Approved by UI Team
 - Able to show feature in company demo
- Attachments:** A file named 'mockup.png' is attached with the description 'Mockup of entry form'.
- Owner:** Greg
- Schedule:**
 - State:** Defined
 - Iteration:** Unscheduled
 - Plan Est:** 8.0 Points
 - To Do:** 0.0 Hours
- Task Est:** 0.0 Hours

At the bottom of the form are four buttons: 'Save & Close', 'Save & New', 'Save', and 'Cancel'.

Stories

Iteration: (any) Release: (any)

Estimated Effort: 36.0 points / Completed: 4.0

Add Story Delete Schedule...

	ID	Story Name	Topic	Type
☐	S3	Add a product to my shopping cart	Shopping Cart	New Feature
☐	S1	View a list of products for sale	Catalog	New Feature
☐	S2	View details of a single product	Catalog	New Feature
☐	S7	Allow user to add a product review	Reviews	New Feature

Stories

Iteration: (any) Release: (any) Type: (any) Status:

Estimated Effort: 36.0 points / Completed: 4.0 points / Remaining: 32.0 points

Add Story Delete Schedule...  Import Stories from Excel

	ID	Story Name	Topic	Type	Pri.	Value	Risk	Estimate
☐	S3	Add a product to my shopping cart	Shopping Cart	New Feature	5			
☐	S1	View a list of products for sale	Catalog	New Feature	5			
☐	S2	View details of a single product	Catalog	New Feature	5			
☐	S7	Allow user to add a product review	Reviews	New Feature	5			
☐	S5	Show high resolution product images	Catalog	New Feature	5			

Import Stories - Step 2: Map Destination Fields

Destination Field	Source Field	Default
Name	Name	(required)
Description	Description	-
Topic		-
Priority		5 - Medium
Estimated Hours	Estimate	0
Requested By		-
Reference		-

Part B:

1. Document the User Stories of selected application.

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Date of performsnce: 21/07/26	DATE OF SUBMISSION: 21/07/26

1)Reception :

- As a receptionist, I want to register new members so they can access new gym services.
- As a receptionist, I want to verify member details so they can be accurate .
- As a receptionist, I want to schedule trials session so new potential customers can become members .
- As a receptionist, I want to print payment receipts so that members receive confirmation of they membership.
- As a receptionist, I want to check the members attendance so I can follow up to them.

2) Trainer Department

- As a trainer, I want to assign workout plans so that members achieve their fitness goals.
- As a trainer, I want to record member progress so that improvements can be monitored.
- As a trainer, I want to schedule personal training sessions so that members receive guidance.
- As a trainer, I want to update exercise routines so that workouts remain effective. As
- a trainer, I want to view member medical restrictions so that training is safe.

3) Administration

- As an administrator, I want to approve new staff accounts so that only authorized employees can access the system.
- As an administrator, I want to manage gym branches so that all locations can be monitored from one system.
- As an administrator, I want to maintain equipment maintenance schedules so that all machines remain operational.
- As an administrator, I want to monitor daily gym attendance so that I can analyze member activity.
- As an administrator, I want to monitor all department activities so that the gym operates efficiently.

4) Membership Management:

- 1. As a membership manager, I want to create membership plans so that the customer can choose suitable packages.
- 2. As a membership manager, I want to cancel the inactive memberships so the accounts are updated.
- 3. As a membership manager, I want to renew memberships so that members can have uninterrupted access.
- 4. As a membership manager, I want to sent renewal reminders so that the members renew on time. 5.
- As a membership manager, I want to track membership status so that I know active and expired members.

5) Accounting:

- 1. As an accounting executive, I want to collect membership payments so that accounts remain updated.
- 2. As an accounting executive, I want to generate invoices so that members receive payment records.
- 3. As an accounting executive, I want to revenue reports so that business performance can be analyzed.,
- 4. As an accounting executive, I want to process refunds so that payment issues are resolved.
- 5. As an accounting executive, I want to track overdue paymentsso that pending dues are collected.

Accepatnce 1:

- 1. Reception

Story 1: Register New Members

Given valid member info (Name, Email, Phone) is entered,

When the receptionist submits the form,

Then create the profile, generate a Member ID, and send a welcome email.

Story 2: Verify Member Details

Given a member provides their Name or ID,

When the receptionist updates any outdated info and saves,

Then update the profile instantly and show a success message.

Story 3: Schedule Trial Sessions

Given an open time slot and a potential customer's details,

When the receptionist books the slot,

Then reserve the trial session and send a confirmation to the customer.

Story 4: Print Payment Receipts

Given a payment is successful,

When the receptionist clicks "Print Receipt",

Then print a receipt with the payment details.

Story 5: Check Member Attendance

Given the reception attendance page is open,

When filtering by date or inactive members,

Then show total visits and a list of members who haven't visited recently.

2. Trainer Department

Story 1: Assign Workout Plans

Given a trainer selects a member's profile,

When the trainer chooses or creates a workout plan and saves,

Then link the plan to the member's profile and notify them.

Story 2: Record Member Progress

Given a member completes a progress test,

When the trainer logs weight or measurements,

Then save the data and update the progress chart.

Story 3: Schedule Personal Training Sessions

Given both trainer and member are free at a specific time,

When the trainer books the session date and time,

Then reserve the slot and add it to both calendars.

Story 4: Update Exercise Routines

Given a member already has a workout plan,

When the trainer changes any exercises, sets, or reps,

Then update the plan and inform the member.

Story 5: View Member Medical Restrictions

Given a member has a health condition logged,

When the trainer opens the member's profile,

Then display a clear medical alert before workouts are set.

3. Administration

Story 1: Approve New Staff Accounts

Given a new staff member registers for access,

When the admin reviews and clicks "Approve",

Then activate the account and send login details.

Story 2: Manage Gym Branches

Given the admin is on the branch management page,

When the admin adds, updates, or disables a branch,

Then update the system records for that location.

Story 3: Maintain Equipment Maintenance Schedules

Given a machine needs regular service,

When the admin sets a recurring service schedule (e.g., every 3 months),

Then set reminders and mark the machine "In Maintenance" during service.

Story 4: Monitor Daily Gym Attendance

Given check-in data is recorded at the entrance,

When the admin views the daily attendance page,

Then display total visits and peak busy hours.

Story 5: Monitor All Department Activities

Given the admin dashboard is open,

When the admin checks system activities,

Then show summary updates from all departments.

4. Membership Management

Story 1: Create Membership Plans

Given price, duration, and access rules are entered,

When the manager saves and publishes the plan, Then
show the plan as available for purchase.

Story 2: Cancel Inactive Memberships

Given a membership has expired or a cancellation request is made, When
the manager confirms cancellation,

Then set status to "Cancelled" and stop gym access.

Story 3: Renew Memberships

Given an expiring or expired profile,

When a renewal payment is completed,

Then extend the expiry date and restore gym access.

Story 4: Send Renewal Reminders

Given a membership is expiring soon (e.g., in 7 days),

When the automated system checks expiring plans, Then
send a reminder email or SMS to the member.

Story 5: Track Membership Status

Given the member directory list, When

filtering by "Active" or "Expired", Then
display the matching list of members.

5. Accounting

Story 1: Collect Membership Payments

Given a member has a balance due, When
payment is made via cash, card, or online, Then
clear the balance and issue a receipt.

Story 2: Generate Invoices

Given a payment or subscription charge,

When billing triggers or is run manually,

Then create a PDF invoice and email it to the member.

Story 3: Generate Revenue Reports

Given system transaction logs,

When the accountant selects a date range,

Then show total income, refunds, and pending payments.

Story 4: Process Refunds

Given an approved refund request,

When the accountant enters the refund details,

Then send funds back to the member and log the transaction.

Story 5: Track Overdue Payments

Given members with late payments,

When opening the overdue accounts page,
Then list unpaid balances and offer an option to send payment notices.